Math 407 Section 1
Fall 2025
Exam 2
March 12, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): ____________________

Student ID: ____________________

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- If the problem asks for a proof, be sure to carefully justify your work, including any theorems from class.

  Note: you may NOT use a theorem or result from class to prove something when it makes the problem entirely trivial. If you are unsure whether a particular theorem or result is allowed, just ask!

Do not write in the table to the right.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

   (a) The number 2 is a prime element of the ring $\mathbb{Z}[\sqrt{-3}]$

   (b) There are exactly five Abelian groups of order 16 up to isomorphism

   (c) In a ring R, if $ab = 0$ then $a = 0$ or $b = 0$

   (d) There are two non-isomorphic non-Abelian groups of order 8

   (e) The element $1 + \sqrt{-3}$ is a unit in $\mathbb{Z}[\sqrt{-3}]$

2. (10 points)

   (a) State the definition of an integral domain

   (b) State the Third Isomorphism Theorem for groups

   (c) State Cayley's Theorem

3. (10 points)

(a) Write down the definition of an irreducible element of an integral domain R

(b) Prove that $6+i$ is an irreducible element of the Gaussian integers $\mathbb{Z}[i]$

4. (10 points)

Let N be a normal subgroup of G with $[G : N] = n$.

(a) State Lagrange's Theorem

(b) Prove that for all $x \in G$, the order of xN in G/N divides n

(c) Prove that $x^n \in N$ for all $x \in G$.

5. (10 points)

The **ring of power series** $\mathbb{R}[[x]]$ is the set of all formal series expressions $\sum_{n=0}^{\infty} a_n x^n$, where $a_0, a_1, a_2, \ldots$ are real numbers, with addition given by

$$\left(\sum_{n=0}^{\infty} a_n x^n\right) + \left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty} (a_n + b_n) x^n$$

and multiplication given by

$$\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty} c_n x^n,$$

where here $c_n = a_n b_0 + a_{n-1} b_1 + \cdots + a_1 b_{n-1} + a_0 b_n$.

(a) Show that in this ring $1 - x$ is a unit. [Hint: geometric series $1 + x + x^2 + \cdots = \frac{1}{1-x}$]

(b) Show that in this ring x is not a unit.

6. (10 points) Let G be a finite group. The **exponent** of G, denoted $\exp(G)$, is the maximum of the orders of elements of G.

   (a) State the Structure Theorem for Finite Abelian Groups in Invariant Factor Form

   (b) If G is a finite Abelian group with invariant factors $a_1, a_2, \ldots, a_r$ (in increasing order), prove that a_r is the exponent of G

   (c) Determine $\exp(G)$ for $G = \mathbb{Z}_{18} \times \mathbb{Z}_{21}$. [Hint: put it in invariant factor form]

7. (10 points)

Let G be a group, suppose $g \in G$ is an element of order n, and let H be the subgroup of G generated by g. Consider the group homomorphism

$$\varphi : \mathbb{Z} \to G, \quad f(k) = g^k.$$

(a) State the First Isomorphism Theorem for groups

(b) Determine the kernel of φ

(c) Explain why $\mathbb{Z}/\ker(\varphi)$ is isomorphic to H